1.12 Coordinate Geometry: Circles

Worked Solutions

1 a)

Centre $(2, 7)$, Radius $= 3$.

1 b)

Centre $(0, -2)$, Radius $= 3.5$.

1 c)

Centre $(-8, 5)$, Radius $= \sqrt{16} = 4$.

1 d)

Centre $(-12, -9)$, Radius $= \sqrt{19}$.

1 e)

Centre $(3, -16)$, Radius $= \sqrt{28} = 2\sqrt{7}$.

1 f)

Centre $(\frac{1}{2}, 0)$, Radius $= \sqrt{\frac{9}{4}} = \frac{3}{2}$.

1 g)

Centre $(-2k, k - 1)$, Radius $= \sqrt{100k} = 10\sqrt{k}$.

2 a)

$(x + 3)^2 + (y + 8)^2 = 9^2$

$\Rightarrow (x + 3)^2 + (y + 8)^2 = 81.$

2 b)

$(x - 4)^2 + (y + 3)^2 = 5^2$

$\Rightarrow (x - 4)^2 + (y + 3)^2 = 25.$

2 c)

$x^2 + y^2 = 100^2$

$\Rightarrow x^2 + y^2 = 10\,000.$

2 d)

$(x - 6)^2 + (y - 1)^2 = (\sqrt{17})^2$

$\Rightarrow (x - 6)^2 + (y - 1)^2 = 17.$

2 e)

$(x + 11)^2 + (y - 6)^2 = (2\sqrt{2})^2$

$\Rightarrow (x + 11)^2 + (y - 6)^2 = 8.$

2 f)

$(x + \frac{2}{3})^2 + (y + \frac{3}{2})^2 = (\frac{4}{9})^2$

$\Rightarrow (x + \frac{2}{3})^2 + (y + \frac{3}{2})^2 = \frac{16}{81}$.

2 g)

$(x - (k - 5))^2 + (y - (7 - k))^2 = (k + 2)^2$

$\Rightarrow (x - k + 5)^2 + (y - 7 + k)^2 = k^2 + 4k + 4$.

3 a)

Radius $= 5$.

Area $= \pi \times 5^2 = 25\pi$ units^2.

3 b)

Radius $= \sqrt{\frac{49}{9}} = \frac{7}{3}$.

Area $= \pi \times (\frac{7}{3})^2 = \frac{49\pi}{9}$ units^2.

3 c)

Radius $= \sqrt{121} = 11$.

Area $= \pi \times 11^2 = 121\pi$ units^2.

4 a)

$x^2 + (y + 6)^2 = 2$

$\Rightarrow x^2 + y^2 + 12y + 36 = 2$

$\Rightarrow x^2 + y^2 + 12y + 34 = 0$.

4 b)

$(x - 10)^2 + y^2 = 130$

$\Rightarrow x^2 - 20x + 100 + y^2 = 130$

$\Rightarrow x^2 - 20x + y^2 - 30 = 0$.

4 c)

$(x + 2)^2 + (y - 8)^2 = 66$

$\Rightarrow x^2 + 4x + 4 + y^2 - 16y + 64 = 66$

$\Rightarrow x^2 + 4x + y^2 - 16y + 2 = 0$.

4 d)

$(x - 11)^2 + (y - 15)^2 = 1600$

$\Rightarrow x^2 - 22x + 121 + y^2 - 30y + 225 = 1600$

$\Rightarrow x^2 - 22x + y^2 - 30y - 1254 = 0$.

4 e)

$(x+4k)^2+(y-2k)^2=20k^2$

$\Rightarrow x^2+8kx+16k^2+y^2-4ky+4k^2=20k^2$

$\Rightarrow x^2+8kx+y^2-4ky=0.$

5 a)

$x^2+2x+y^2+6y-8=0$

$\Rightarrow (x+1)^2-1+(y+3)^2-9-8=0$

$\Rightarrow (x+1)^2+(y+3)^2=18.$

5 b)

$x^2-8x+y^2-20y-784=0$

$\Rightarrow (x-4)^2-16+(y-10)^2-100-784=0$

$\Rightarrow (x-4)^2+(y-10)^2=900.$

5 c)

$x^2+6x+y^2-14y-82=0$

$\Rightarrow (x+3)^2-9+(y-7)^2-49-82=0$

$\Rightarrow (x+3)^2+(y-7)^2=140.$

5 d)

$x^2-5x+y^2-9y-\frac{57}{2}=0$

$\Rightarrow (x-\frac{5}{2})^2-\frac{25}{4}+(y-\frac{9}{2})^2-\frac{81}{4}-\frac{57}{2}=0$

$\Rightarrow (x-\frac{5}{2})^2+(y-\frac{9}{2})^2=55.$

5 e)

$x^2-\frac{2}{3}x+y^2+\frac{4}{9}y-\frac{230}{81}=0$

$\Rightarrow (x-\frac{1}{3})^2-\frac{1}{9}+(y+\frac{2}{9})^2-\frac{4}{81}-\frac{230}{81}=0$

$\Rightarrow (x-\frac{1}{3})^2+(y+\frac{2}{9})^2=3.$

5 f)

$x^2-\frac{3}{2}x+y^2-\frac{5}{4}y-\frac{1091}{64}=0$

$\Rightarrow (x-\frac{3}{4})^2-\frac{9}{16}+(y-\frac{5}{8})^2-\frac{25}{64}-\frac{1091}{64}=0$

$\Rightarrow (x-\frac{3}{4})^2+(y-\frac{5}{8})^2=18.$

5 g)

$x^2+\frac{14}{5}x+y^2+\frac{5}{2}y+\frac{1009}{400}=0$

$\Rightarrow (x+\frac{7}{5})^2-\frac{49}{25}+(y+\frac{5}{4})^2-\frac{25}{16}+\frac{1009}{400}=0$

$\Rightarrow (x+\frac{7}{5})^2+(y+\frac{5}{4})^2=1.$

5 h)

$x^2 + kx + y^2 - 6ky + 3k = 0$

$\Rightarrow (x + \frac{1}{2}k)^2 - \frac{1}{4}k^2 + (y - 3k)^2 - 9k^2 + 3k = 0$

$\Rightarrow (x + \frac{1}{2}k)^2 + (y - 3k)^2 = \frac{37}{4}k^2 - 3k.$

6 a)

$x^2 - 6x + y^2 + 16y + 83 = 0$

$\Rightarrow (x - 3)^2 - 9 + (y + 8)^2 - 64 + 83 = 0$

$\Rightarrow (x - 3)^2 + (y + 8)^2 = -10.$

The right-hand side is negative, so this does not represent a circle. There are no real coordinates that satisfy this equation.

6 b)

$x^2 + 10x + y^2 - y + \frac{101}{4} = 0$

$\Rightarrow (x + 5)^2 - 25 + (y - \frac{1}{2})^2 - \frac{1}{4} + \frac{101}{4} = 0$

$\Rightarrow (x + 5)^2 + (y - \frac{1}{2})^2 = 0.$

The right-hand side is zero, so this does not represent a circle. Only one point satisfies the equation, namely $(-5, \frac{1}{2})$.

6 c)

$x^2 - 4x - y^2 - 10y - 46 = 0$

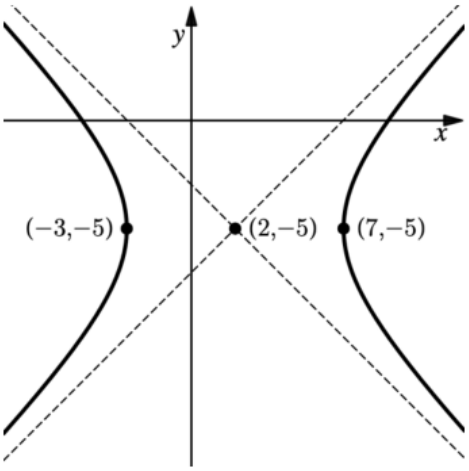
$\Rightarrow (x - 2)^2 - 4 - (y^2 + 10y) - 46 = 0$

$\Rightarrow (x - 2)^2 - 4 - ((y + 5)^2 - 25) - 46 = 0$

$\Rightarrow (x - 2)^2 - 4 - (y + 5)^2 + 25 - 46 = 0$

$\Rightarrow (x - 2)^2 - (y + 5)^2 = 25.$

As the coefficient of y^2 is not equal to 1, this does not represent a circle. In fact, this is a hyperbola, as shown below:



6 d)

$x^2 + 12x + 3y^2 - 6y + 3 = 0$

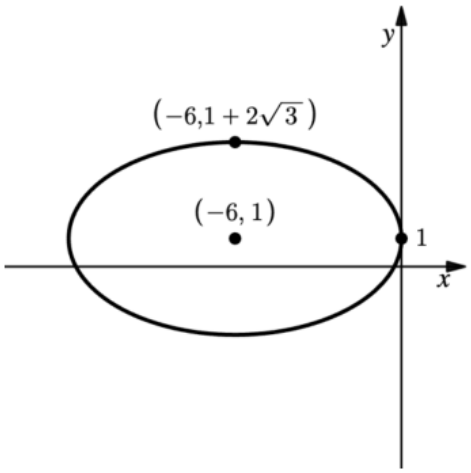
$\Rightarrow (x+6)^2 - 36 + 3(y^2 - 2y) + 3 = 0$

$\Rightarrow (x+6)^2 - 36 + 3((y-1)^2 - 1) + 3 = 0$

$\Rightarrow (x+6)^2 - 36 + 3(y-1)^2 - 3 + 3 = 0$

$\Rightarrow (x+6)^2 + 3(y-1)^2 = 36.$

As the coefficient of y^2 is not equal to 1, this does not represent a circle. In fact, this is an ellipse, as shown below:



6 e)

$x^2 + \frac{2}{3}x + y^2 - \frac{4}{3}y - \frac{1}{3} = 0$

$\Rightarrow (x + \frac{1}{3})^2 - \frac{1}{9} + (y - \frac{2}{3})^2 - \frac{4}{9} - \frac{1}{3} = 0$

$\Rightarrow (x + \frac{1}{3})^2 + (y - \frac{2}{3})^2 = \frac{8}{9}.$

This is a circle centred at $(-\frac{1}{3}, \frac{2}{3})$ with radius $\sqrt{\frac{8}{9}} = \frac{2\sqrt{2}}{3}$.

7)

$x^2 + 8x + y^2 - 12y - k = 0$

$\Rightarrow (x+4)^2 - 16 + (y-6)^2 - 36 - k = 0$

$\Rightarrow (x+4)^2 + (y-6)^2 = 52 + k$

$52 + k > 0 \Rightarrow k > -52.$

8)

$x^2 - 3x + y^2 + 9y + \frac{k}{4} = 0$

$\Rightarrow (x - \frac{3}{2})^2 - \frac{9}{4} + (y + \frac{9}{2})^2 - \frac{81}{4} + \frac{k}{6} = 0$

$\Rightarrow (x - \frac{3}{2})^2 + (y + \frac{9}{2})^2 = \frac{45}{2} - \frac{k}{6}$

$\frac{45}{2} - \frac{k}{6} > 0 \Rightarrow k < 135.$

9 a)

$(x-4)^2+(y-5)^2=9$.

This circle has a centre at $(4,5)$ with radius 3.

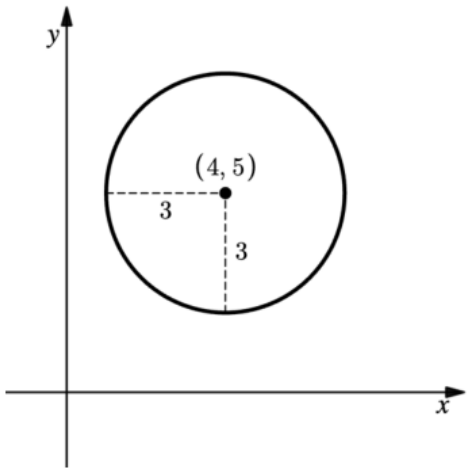
$3<4$ so the circle does not intersect the y-axis.

$3<5$ so the circle does not intersect the x-axis.

Alternatively:

Let $x=0$: $16+(y-5)^2=9 \Rightarrow (y-5)^2=-7$ which has no real solutions, so the circle does not intersect the y-axis.

Let $y=0$: $(x-4)^2+25=9 \Rightarrow (x-4)^2=-16$ which has no real solutions, so the circle does not intersect the x-axis.



9 b)

$(x+1)^2 + (y-6)^2 = 25$.

The circle has a centre at $(-1, 6)$ with radius 5.

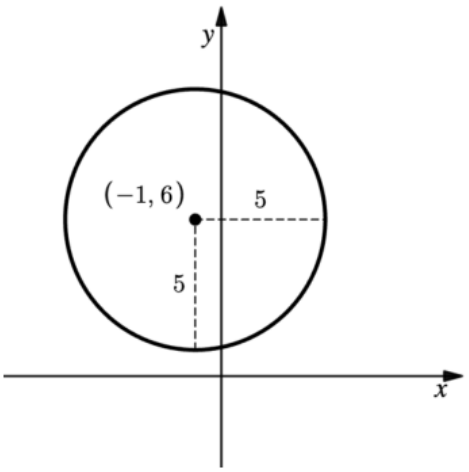
$5 > |-1|$ so the circle intersects the y-axis.

$5 < 6$ so the circle does not intersect the x-axis.

Alternatively:

Let $x = 0$: $1 + (y-6)^2 = 25 \Rightarrow (y-6)^2 = 24$ which has two distinct real solutions, so the circle does intersect the y-axis.

Let $y = 0$: $(x+1)^2 + 36 = 9 \Rightarrow (x+1)^2 = -27$ which has no real solutions, so the circle does not intersect the x-axis.



9 c)

$(x+3)^2+(y-4)^2=8$.

The circle has a centre at $(-3,4)$ with radius $2\sqrt{2}$.

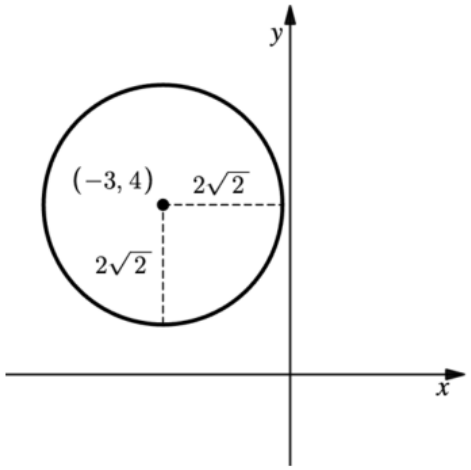
$2\sqrt{2}<|-3|$ so the circle does not intersect the y-axis.

$2\sqrt{2}<4$ so the circle does not intersect the x-axis.

Alternatively:

Let $x=0$: $9+(y-4)^2=8 \Rightarrow (y-4)^2=-1$ which has no real solutions, so the circle does not intersect the y-axis.

Let $y=0$: $(x+3)^2+16=8 \Rightarrow (x+3)^2=-8$ which has no real solutions, so the circle does not intersect the x-axis.



9 d)

$(x+8)^2+(y+3)^2=10$.

The circle has a centre at $(-8,-3)$ with radius $\sqrt{10}$.

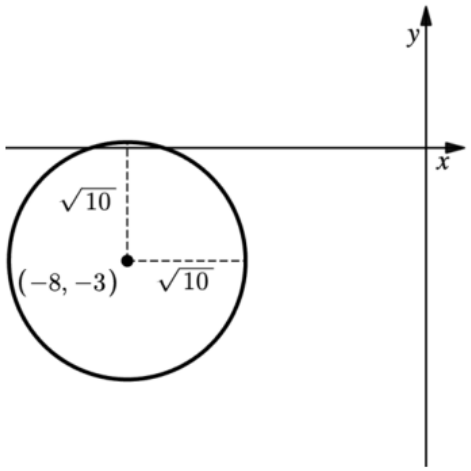
$\sqrt{10}<|-8|$ so the circle does not intersect the y-axis.

$\sqrt{10}>3$ so the circle intersects the x-axis.

Alternatively:

Let $x=0$: $64+(y+3)^2=10 \Rightarrow (y+3)^2=-54$ which has no real solutions, so the circle does not intersect the y-axis.

Let $y=0$: $(x+8)^2+9=10 \Rightarrow (x+8)^2=1$ which has two distinct real solutions, so the circle does intersect the x-axis.



10 a)

$(x-2)^2+(y-k)^2=36$.

The circle has a centre at $(2,k)$ with radius 6.

The circle will intersect the x-axis at two distinct points when $0<k<6$.

10 b)

$(x+5)^2+(y-k)^2=10$.

The circle has a centre at $(-5,k)$ with radius $\sqrt{10}$.

The circle will intersect the x-axis at two distinct points when $0<k<\sqrt{10}$.

10 c)

$(x-1)^2+(y-4)^2=k^2$.

The circle has a centre at $(1,4)$ with radius k.

The circle will intersect the x-axis at two distinct points when $k>4$.

10 d)

$(x+7)^2+(y+8)^2=(k+1)^2$.

The circle has a centre at $(-7,-8)$ with radius $k+1$. The circle will intersect the x-axis at two distinct points when $k+1>8$

$\Rightarrow k>7$.

11 a)

$(x-k)^2+y^2=49$.

The circle has a centre at $(k,0)$ with radius 7.

The circle will intersect the y-axis at two distinct points when $0<k<7$.

11 b)

$(x+k)^2+(y-4)^2=9$.

The circle has a centre at $(-k,4)$ with radius 3.

The circle will intersect the y-axis at two distinct points when $0<k<3$.

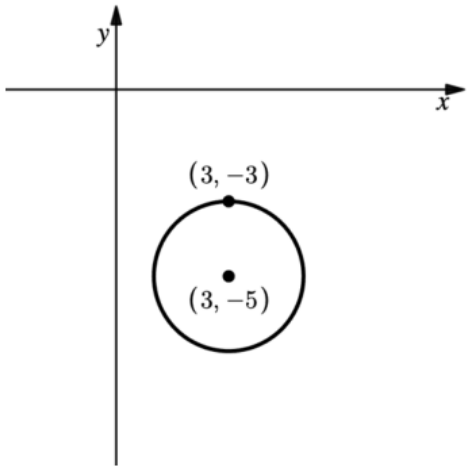
11 c)

$(x-6)^2+(y+3)^2=k^2$.

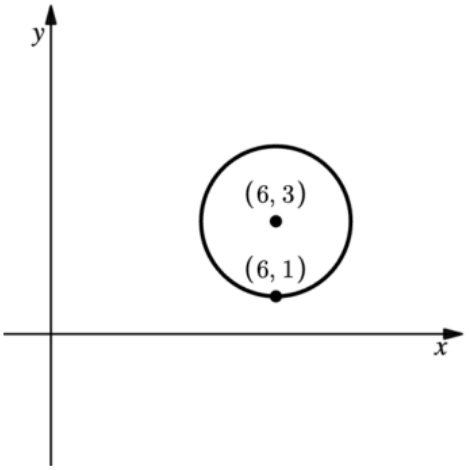
The circle has a centre at $(6,-3)$ with radius k.

The circle will intersect the y-axis at two distinct points when $k>6$.

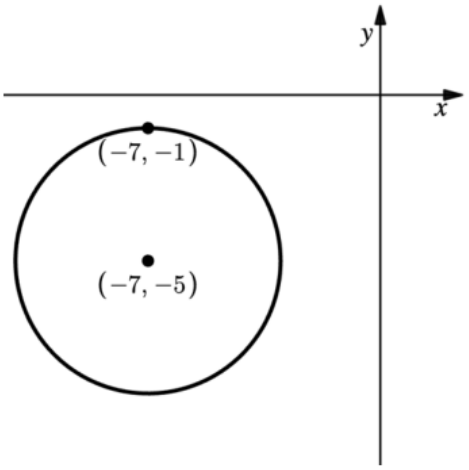
12 a)



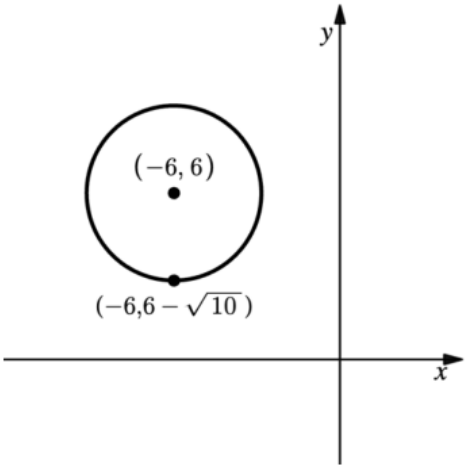
12 b)



12 c)



12 d)



13 a)

$x^2 + (y-3)^2 = 9$.

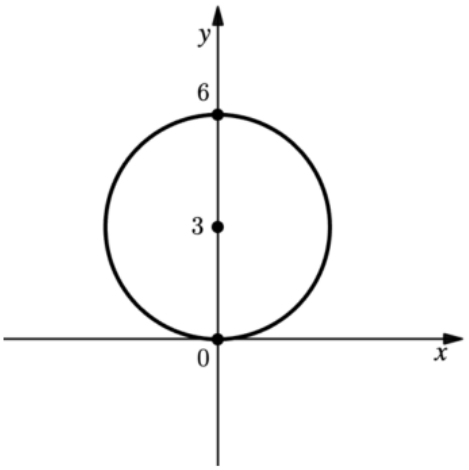
When $x = 0$: $(y-3)^2 = 9 \Rightarrow y - 3 = \pm 3$

$\Rightarrow y = \pm 3 + 3 = 6$ or 0.

When $y = 0$: $x^2 + (-3)^2 = 9 \Rightarrow x^2 + 9 = 9$

$\Rightarrow x^2 = 0 \Rightarrow x = 0$.

So the circle intersects the coordinate axes at $(0, 6)$ and $(0, 0)$.



13 b)

$x^2 + (y+2)^2 = 16$.

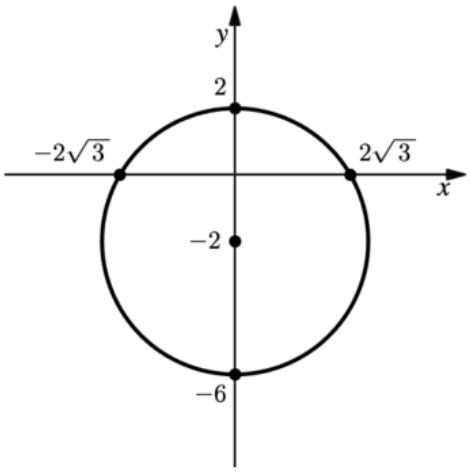
When $x = 0$: $(y+2)^2 = 16 \Rightarrow y + 2 = \pm 4$

$\Rightarrow y = \pm 4 - 2 = 2$ or -6.

When $y = 0$: $x^2 + 2^2 = 16 \Rightarrow x^2 + 4 = 16$

$\Rightarrow x^2 = 12 \Rightarrow x = \pm 2\sqrt{3}$.

So the circle intersects the coordinate axes at $(0, 2)$, $(0, -6)$, $(-2\sqrt{3}, 0)$ and $(2\sqrt{3}, 0)$.



13 c)

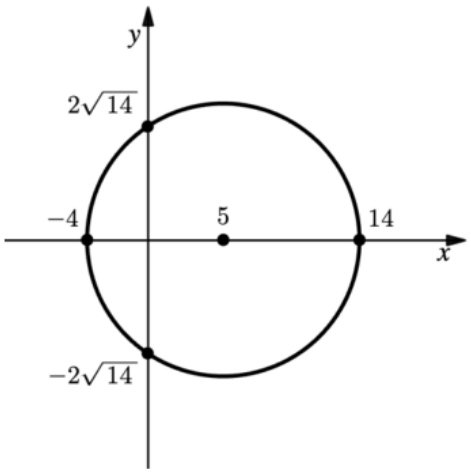
$(x-5)^2+y^2=81$.

When $x=0$: $(-5)^2+y^2=81 \Rightarrow 25+y^2=81 \Rightarrow y^2=56 \Rightarrow y=\pm 2\sqrt{14}$.

When $y=0$: $(x-5)^2=81 \Rightarrow x-5=\pm 9$

$\Rightarrow x=\pm 9+5 \Rightarrow x=14$ or -4.

So the circle intersects the coordinate axes at $(0,-2\sqrt{14})$, $(0,2\sqrt{14})$, $(-4,0)$ and $(14,0)$.



13 d)

$(x+5)^2+y^2=25$.

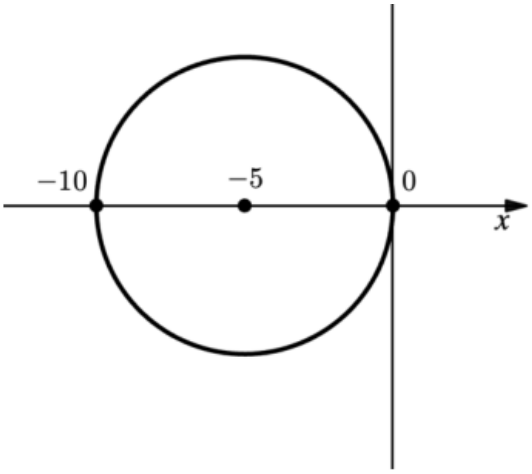
When $x=0$: $5^2+y^2=25 \Rightarrow 25+y^2=25$

$\Rightarrow y^2=0 \Rightarrow y=0$.

When $y=0$: $(x+5)^2=25 \Rightarrow x+5=\pm 5$

$\Rightarrow x=\pm 5-5 \Rightarrow x=0$ or -10.

So the circle intersects the coordinate axes at $(-10,0)$ and $(0,0)$.



14 a)

$(x-4)^2+(y-4)^2=16$.

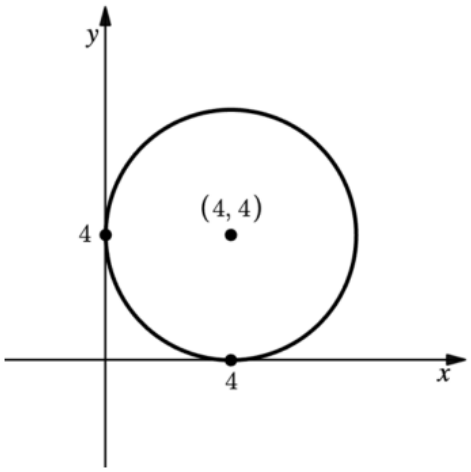
When $x=0$: $(-4)^2+(y-4)^2=16$

$\Rightarrow 16+(y-4)^2=16 \Rightarrow (y-4)^2=0 \Rightarrow y=4$.

When $y=0$: $(x-4)^2+(-4)^2=16$

$\Rightarrow (x-4)^2+16=16 \Rightarrow (x-4)^2=0 \Rightarrow x=4$.

So the circle intersects the coordinate axes at $(0,4)$ and $(4,0)$.



14 b)

$(x+2)^2+(y+2)^2=13$.

When $x=0$: $2^2+(y+2)^2=13$

$\Rightarrow 4+(y+2)^2=13 \Rightarrow (y+2)^2=9$

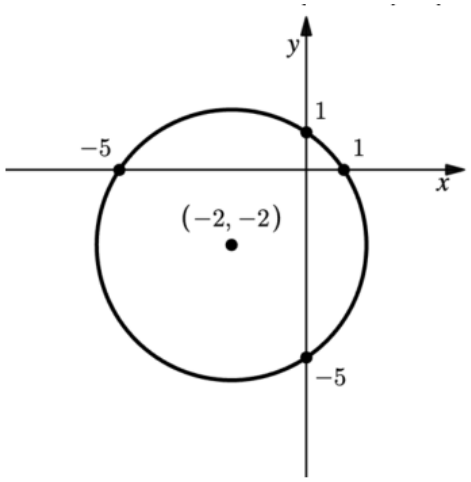
$\Rightarrow y+2=\pm 3 \Rightarrow y=\pm 3-2=1$ or -5.

When $y=0$: $(x+2)^2+2^2=13$

$\Rightarrow (x+2)^2+4=13 \Rightarrow (x+2)^2=9$

$\Rightarrow x+2=\pm 3 \Rightarrow x=\pm 3-2=1$ or -5.

So the circle intersects the coordinate axes at $(0,-5)$, $(0,1)$, $(-5,0)$ and $(1,0)$.



14 c)

$(x-1)^2+(y+3)^2=17$.

When $x=0$: $(-1)^2+(y+3)^2=17$

$\Rightarrow 1+(y+3)^2=17 \Rightarrow (y+3)^2=16$

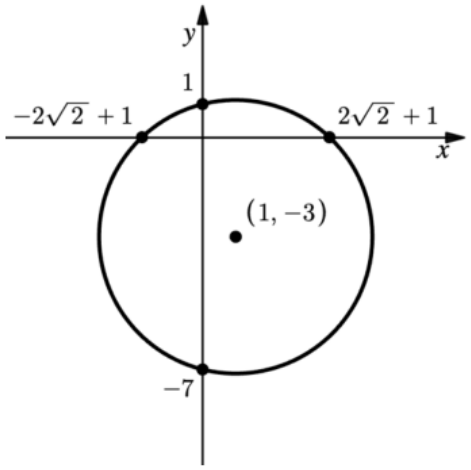
$\Rightarrow y+3=\pm 4 \Rightarrow y=\pm 4-3=1$ or -7.

When $y=0$: $(x-1)^2+3^2=17$

$\Rightarrow (x-1)^2+9=17 \Rightarrow (x-1)^2=8$

$\Rightarrow x-1=\pm 2\sqrt{2} \Rightarrow x=\pm 2\sqrt{2}+1=2\sqrt{2}+1$ or $-2\sqrt{2}+1$.

So the circle intersects the coordinate axes at $(0,-7)$, $(0,1)$, $(-2\sqrt{2}+1,0)$ and $(2\sqrt{2}+1,0)$.



14 d)

$(x+7)^2 + (y-2)^2 = 20$.

When $x = 0$: $7^2 + (y-2)^2 = 20$

$\Rightarrow 49 + (y-2)^2 = 20 \Rightarrow (y-2)^2 = -29$

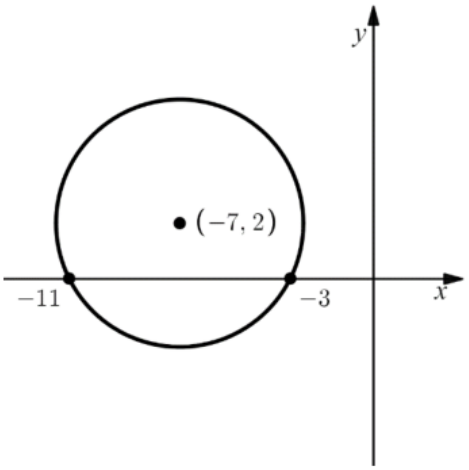
$\Rightarrow$ No real solutions, so the circle does not intersect the y-axis.

When $y = 0$: $(x+7)^2 + (-2)^2 = 20$

$\Rightarrow (x+7)^2 + 4 = 20 \Rightarrow (x+7)^2 = 16$

$\Rightarrow x + 7 = \pm 4 \Rightarrow x = -11$ or -3.

So the circle intersects the coordinate axes at $(-11, 0)$ and $(-3, 0)$.



15 a)

$(x+3)^2 + (y-8)^2 = 16$.

When $x = 0$: $3^2 + (y-8)^2 = 16$

$\Rightarrow 9 + (y-8)^2 = 16 \Rightarrow (y-8)^2 = 7$

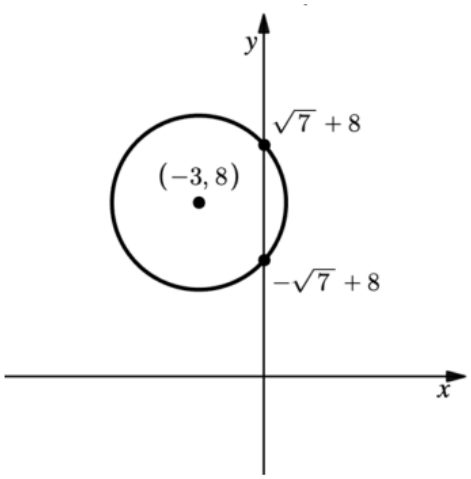
$\Rightarrow y - 8 = \pm\sqrt{7} \Rightarrow y = \pm\sqrt{7} + 8 = \sqrt{7} + 8$ or $-\sqrt{7} + 8$.

When $y = 0$: $(x+3)^2 + (-8)^2 = 16$

$\Rightarrow (x+3)^2 + 64 = 16 \Rightarrow (x+3)^2 = -48$

$\Rightarrow$ No real solutions, so the circle does not intersect the x-axis.

So the circle intersects the coordinate axes at $(0, -\sqrt{7} + 8)$ and $(0, \sqrt{7} + 8)$.



15 b)

$(x+7)^2+(y+5)^2=36$.

When $x=0$: $7^2+(y+5)^2=36$

$\Rightarrow 49+(y+5)^2=36 \Rightarrow (y+5)^2=-13$

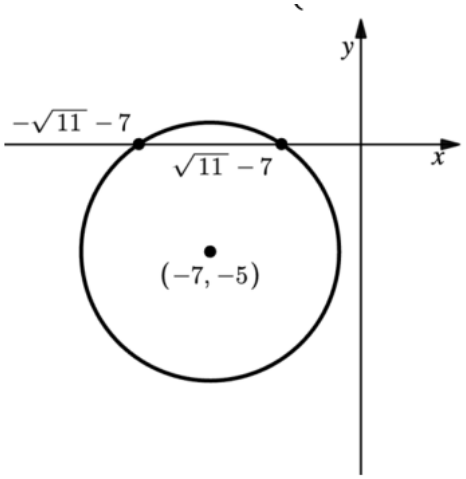
$\Rightarrow$ No real solutions, so the circle does not intersect the y-axis.

When $y=0$: $(x+7)^2+5^2=36$

$\Rightarrow (x+7)^2+25=36 \Rightarrow (x+7)^2=11$

$\Rightarrow x+7=\pm\sqrt{11} \Rightarrow x=\pm\sqrt{11}-7=\sqrt{11}-7$ or $-\sqrt{11}-7$.

So the circle intersects the coordinate axes at $(-\sqrt{11}-7,0)$ and $(\sqrt{11}-7,0)$.



15 c)

$(x-8)^2+(y-2)^2=81$.

When $x=0$: $(-8)^2+(y-2)^2=81$

$\Rightarrow 64+(y-2)^2=81 \Rightarrow (y-2)^2=17$

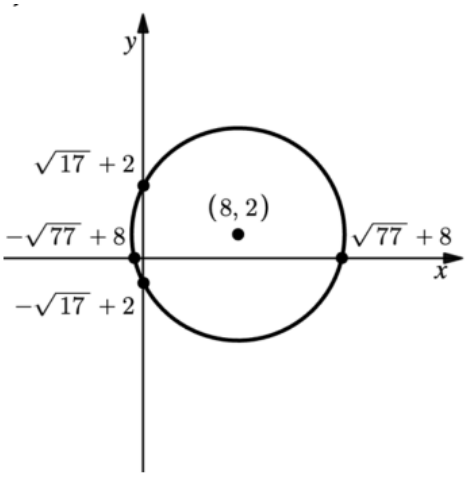
$\Rightarrow y-2=\pm\sqrt{17} \Rightarrow y=\pm\sqrt{17}+2=\sqrt{17}+2$ or $-\sqrt{17}+2$.

When $y=0$: $(x-8)^2+(-2)^2=81$

$\Rightarrow (x-8)^2+4=81 \Rightarrow (x-8)^2=77$

$\Rightarrow x-8=\pm\sqrt{77} \Rightarrow y=\pm\sqrt{77}+8=\sqrt{77}+8$ or $-\sqrt{77}+8$.

So the circle intersects the coordinate axes at $(0,-\sqrt{17}+2)$, $(0,\sqrt{17}+2)$, $(-\sqrt{77}+8,0)$ and $(\sqrt{77}+8,0)$.



15 d)

$(x-1)^2+(y-\frac{11}{16})^2=\frac{697}{256}$.

When $x=0$: $(-1)^2+(y-\frac{11}{16})^2=\frac{697}{256}$

$\Rightarrow 1+(y-\frac{11}{16})^2=\frac{697}{256} \Rightarrow (y-\frac{11}{16})^2=\frac{441}{256}$

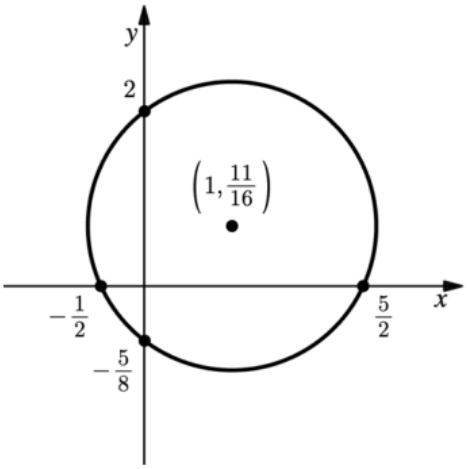
$\Rightarrow y-\frac{11}{16}=\pm\frac{21}{16} \Rightarrow y=\pm\frac{21}{16}+\frac{11}{16}=2$ or $-\frac{5}{8}$.

When $y=0$: $(x-1)^2+(-\frac{11}{16})^2=\frac{697}{256}$

$\Rightarrow (x-1)^2+\frac{121}{256}=\frac{697}{256} \Rightarrow (x-1)^2=\frac{9}{4}$

$\Rightarrow x-1=\pm\frac{3}{2} \Rightarrow y=\pm\frac{3}{2}+1=\frac{5}{2}$ or $-\frac{1}{2}$.

So the circle intersects the coordinate axes at $(0,-\frac{5}{8})$, $(0,2)$, $(-\frac{1}{2},0)$ and $(\frac{5}{2},0)$.



16 a)

$(3-1)^2+(7-5)^2=8<9$ so $(3,7)$ is inside the circle.

16 b)

$(3-3)^2+(7-9)^2=4$ so $(3,7)$ is on the circle.

16 c)

$(3-5)^2+(7-6)^2=5>1$ so $(3,7)$ is outside the circle.

17)

$r=\sqrt{(14-2)^2+(14+2)^2}=20$.

$\therefore (x-2)^2+(y+2)^2=400$.

18 a)

Centre $=(\frac{-5+9}{2},\frac{-7-5}{2})=(2,-6)$.

Distance from $A(-5,-7)$ to centre:

$r=\sqrt{(2+5)^2+(-6+7)^2}=\sqrt{50}$.

$\therefore (x-2)^2+(y+6)^2=50$.

18 b)

Centre $= (\frac{-8+6}{2}, \frac{7-1}{2}) = (-1, 3)$.

Distance from $A(-8, 7)$ to centre:

$r = \sqrt{(-8+1)^2 + (7-3)^2} = \sqrt{65}$.

$\therefore (x+1)^2 + (y-3)^2 = 65$.

18 c)

Centre $= (\frac{4-4}{2}, \frac{7-9}{2}) = (0, -1)$.

Distance from $A(4, 7)$ to centre:

$r = \sqrt{(4-0)^2 + (7+1)^2} = \sqrt{64}$.

$\therefore x^2 + (y+1)^2 = 80$.

18 d)

Centre $= (\frac{-2+12}{2}, \frac{-1+13}{2}) = (5, 6)$.

Distance from $A(-2, -1)$ to centre:

$r = \sqrt{(-2-5)^2 + (-1-6)^2} = \sqrt{98}$.

$\therefore (x-5)^2 + (y-6)^2 = 98$.

19)

Midpoint of A and $B = (\frac{6+8}{2}, \frac{2+4}{2}) = (7, 3)$.

Gradient of $AB = \frac{4-2}{8-6} = \frac{2}{2} = 1$.

Perpendicular Bisector: $y - 3 = -(x - 7)$

$\Rightarrow y = -x + 10$.

When $y = 0$: $0 = -x + 10 \Rightarrow x = 10$.

Centre of circle is $(10, 0)$.

$r = \sqrt{(6-10)^2 + (2-0)^2} = \sqrt{20}$.

$\therefore (x-10)^2 + y^2 = 20$.

20)

Midpoint of A and $B = (\frac{-5+3}{2}, \frac{1+11}{2}) = (-1, 6)$.

Gradient of $AB = \frac{11-1}{3+5} = \frac{10}{8} = \frac{5}{4}$.

Perpendicular Bisector: $y - 6 = -\frac{4}{5}(x + 1)$

$\Rightarrow y = -\frac{4}{5}x + \frac{26}{5}$.

When $x = 0$: $y = \frac{26}{5}$.

Centre of circle is $(0, \frac{26}{5})$.

$r = \sqrt{(-5-0)^2 + (1-\frac{26}{5})^2} = \sqrt{\frac{1066}{25}}$.

$\therefore x^2 + (y - \frac{26}{5})^2 = \frac{1066}{25}$.

21)

Midpoint of A and $B = (\frac{1+17}{2}, \frac{-6-10}{2}) = (9, -8)$.

Gradient of $AB = \frac{-10+6}{17-1} = \frac{-4}{16} = -\frac{1}{4}$.

Perpendicular Bisector: $y + 8 = 4(x - 9)$

$\Rightarrow y = 4x - 44$.

When $y = x$: $x = 4x - 44 \Rightarrow 3x = 44 \Rightarrow x = \frac{44}{3}$.

Centre of circle is $(\frac{44}{3}, \frac{44}{3})$.

$r = \sqrt{(1 - \frac{44}{3})^2 + (-6 - \frac{44}{3})^2} = \sqrt{\frac{5525}{9}}$.

$\therefore (x - \frac{44}{3})^2 + (y - \frac{44}{3})^2 = \frac{5525}{9}$.

22)

Centre $= C(8, -3)$, Radius $= 3$.

Distance from A to C

$= \sqrt{(8+3)^2 + (-3-8)^2} = \sqrt{242}$.

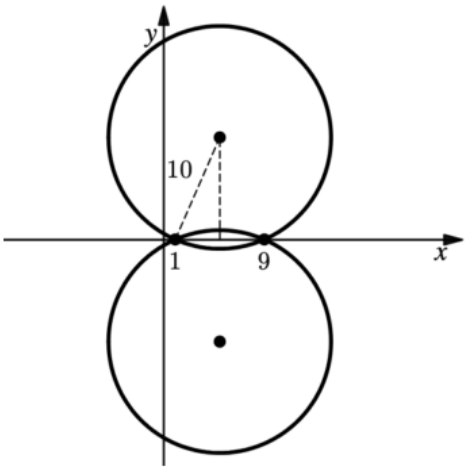
Distance from A to the circumference of the circle is $\sqrt{242} - 3 = 11\sqrt{2} - 3$.

23)

$\sqrt{10^2 - 4^2} = \sqrt{84} = 2\sqrt{21}$.

$(x - 5)^2 + (y - 2\sqrt{21})^2 = 100$

and $(x - 5)^2 + (y + 2\sqrt{21})^2 = 100$.



24)

Midpoint of A and $B = (\frac{7+3}{2}, \frac{2+2}{2}) = (5, 2)$.

Perpendicular Bisector: $x = 5$.

Midpoint of B and $C = (\frac{3+3}{2}, \frac{2-6}{2}) = (3, -2)$.

Perpendicular Bisector: $y = -2$.

Centre of circle is $(5, -2)$.

$r = \sqrt{(7-5)^2 + (2+2)^2} = \sqrt{20}$.

$\therefore (x - 5)^2 + (y + 2)^2 = 20$.

25)

Midpoint of A and $B = (\frac{-1+0}{2}, \frac{-12-13}{2}) = (-\frac{1}{2}, -\frac{25}{2})$.

Gradient of $AB = \frac{-12+13}{-1-0} = \frac{1}{-1} = -1$.

Perpendicular Bisector: $y + \frac{25}{2} = x + \frac{1}{2}$

$\Rightarrow y = x - 12$.

Midpoint of B and $C = (\frac{0+3}{2}, \frac{-13-4}{2}) = (\frac{3}{2}, -\frac{17}{2})$.

Gradient of $BC = \frac{-4+13}{3-0} = \frac{9}{3} = 3$.

Perpendicular Bisector: $y + \frac{17}{2} = -\frac{1}{3}(x - \frac{3}{2})$

$\Rightarrow y = -\frac{1}{3}x - 8$.

$x - 12 = -\frac{1}{3}x - 8 \Rightarrow 3x - 36 = -x - 24$

$\Rightarrow 4x = 12 \Rightarrow x = 3 \Rightarrow y = 3 - 12 = -9$.

Centre of circle is $(3, -9)$.

$r = \sqrt{(3+1)^2 + (-9+12)^2} = \sqrt{25}$.

$\therefore (x-3)^2 + (y+9)^2 = 25$.

26)

Midpoint of A and $B = (\frac{-8-9}{2}, \frac{1+2}{2}) = (-\frac{17}{2}, \frac{3}{2})$.

Gradient of $AB = \frac{2-1}{-9+8} = \frac{1}{-1} = -1$.

Perpendicular Bisector: $y - \frac{3}{2} = x + \frac{17}{2}$

$\Rightarrow y = x + 10$.

Midpoint of B and $C = (\frac{-9+0}{2}, \frac{2+11}{2}) = (-\frac{9}{2}, \frac{13}{2})$.

Gradient of $BC = \frac{11-2}{0+9} = \frac{9}{9} = 1$.

Perpendicular Bisector: $y - \frac{13}{2} = -(x + \frac{9}{2})$

$\Rightarrow y = -x + 2$.

$x + 10 = -x + 2 \Rightarrow 2x = -8 \Rightarrow x = -4$

$\Rightarrow y = -4 + 10 = 6$.

Centre of circle is $(-4, 6)$.

$r = \sqrt{(-8+4)^2 + (1-6)^2} = \sqrt{41}$.

$\therefore (x+4)^2 + (y-6)^2 = 41$.

27)

Midpoint of A and $B = (\frac{1+2}{2}, \frac{1+6}{2}) = (\frac{3}{2}, \frac{7}{2})$.

Gradient of $AB = \frac{6-1}{2-1} = \frac{5}{1} = 5$.

Perpendicular Bisector: $y - \frac{7}{2} = -\frac{1}{5}(x - \frac{3}{2})$

$\Rightarrow y = -\frac{1}{5}x + \frac{19}{5}$.

When $x = 0$: $y = -\frac{1}{5} \times 0 + \frac{19}{5} = \frac{19}{5}$.

So the centre of the circle is at $(0, \frac{19}{5})$.

$r = \sqrt{(1-0)^2 + (1-\frac{19}{5})^2} = \sqrt{\frac{221}{25}}$.

$\therefore x^2 + (y - \frac{19}{5})^2 = \frac{221}{25}$.

28)

As AC is a diameter of the circle, the triangle ABC is right-angled with AC as the hypotenuse.

$AB = \sqrt{(2+6)^2 + (7+1)^2} = 8\sqrt{2}$.

$BC = \sqrt{(4-2)^2 + (5-7)^2} = 2\sqrt{2}$.

Area $= \frac{1}{2} \times 8\sqrt{2} \times 2\sqrt{2} = 16$ units^2.

29 a)

$AB = \sqrt{(-8+12)^2 + (-1-11)^2} = \sqrt{160}$.

$BC = \sqrt{(-2+8)^2 + (1+1)^2} = \sqrt{40}$.

$AC = \sqrt{(-2+12)^2 + (1-11)^2} = \sqrt{200}$.

29 b)

If AC is a diameter of the circle, then $\angle ABC = 90°$.

If Pythagoras' Theorem holds then this is true.

$AB^2 + BC^2 = (\sqrt{160})^2 + (\sqrt{40})^2 = 160 + 40$

$= 200 = (\sqrt{200})^2 = AC^2$.

$\therefore AC$ is the hypotenuse of the triangle ABC, and so AC is a diameter of the circle.

29 c)

Area $= \frac{1}{2} \times \sqrt{160} \times \sqrt{40} = 40$ units^2.

30)

The line through $(2, 6)$ and $(10, 20)$:

$m = \frac{20-6}{10-2} = \frac{14}{8} = \frac{7}{4}$.

$y - 6 = \frac{7}{4}(x - 2)$.

Sub in $(6, k)$: $k - 6 = \frac{7}{4}(6 - 2) \Rightarrow k = 13$.

Centre $= C(\frac{2+6}{2}, \frac{6+13}{2}) = C(4, \frac{19}{2})$.

$AC = \sqrt{(4-2)^2 + (\frac{19}{2} - 6)^2} = \sqrt{\frac{65}{4}}$.

$\therefore (x-4)^2 + (y - \frac{19}{2})^2 = \frac{65}{4}$.

31 a)

$(x+4)^2 + (5x + 1 - 6)^2 = 20$

$\Rightarrow (x+4)^2 + (5x - 5)^2 = 20$

$\Rightarrow x^2 + 8x + 16 + 25x^2 - 50x + 25 = 20$

$\Rightarrow 26x^2 - 42x + 21 = 0$.

$b^2 - 4ac = (-42)^2 - 4 \times 26 \times 21 = -420 < 0$.

So the line does not intersect the circle.

31 b)

$(x+4)^2 + (1 - 4x - 6)^2 = 20$

$\Rightarrow (x+4)^2 + (-4x - 5)^2 = 20$

$\Rightarrow x^2 + 8x + 16 + 16x^2 + 40x + 25 = 20$

$\Rightarrow 17x^2 + 48x + 21 = 0.$

$b^2 - 4ac = 48^2 - 4 \times 17 \times 21 = 876 > 0.$

So the line intersects the circle at two distinct points.

31 c)

$(x + 4)^2 + (2x + 24 - 6)^2 = 20$

$\Rightarrow (x + 4)^2 + (2x + 18)^2 = 20$

$\Rightarrow x^2 + 8x + 16 + 4x^2 + 72x + 324 = 20$

$\Rightarrow 5x^2 + 80x + 320 = 0.$

$b^2 - 4ac = 80^2 - 4 \times 5 \times 320 = 0.$

So the line is a tangent to the circle.

32)

$x^2 - 3x + y^2 + 10y + \frac{427}{16} = 0$

$\Rightarrow (x - \frac{3}{2})^2 - \frac{9}{4} + (y + 5)^2 - 25 + \frac{427}{16} = 0$

$\Rightarrow (x - \frac{3}{2})^2 + (y + 5)^2 = \frac{9}{16}.$

Centre $= (\frac{3}{2}, -5)$, Radius $= \frac{3}{4}$.

$y = -5 \pm \frac{3}{4}$ so $y = -\frac{17}{4}$ and $y = -\frac{23}{4}$ are tangents.

33)

Let $O(0, 0)$, $A(-10, 10\sqrt{3})$ and $B(10, 10\sqrt{3})$.

Midpoint of O and $A = (\frac{0-10}{2}, \frac{0+10\sqrt{3}}{2}) = (-5, 5\sqrt{3})$.

Gradient of $OA = \frac{10\sqrt{3}-0}{-10-0} = -\sqrt{3}$.

Perpendicular bisector of OA:

$y - 5\sqrt{3} = \dfrac{1}{\sqrt{3}}(x + 5) \Rightarrow y = \frac{\sqrt{3}}{3}x + \frac{20\sqrt{3}}{3}.$

Triangle is symmetrical about the y-axis, so the centre of the circle lies on the y-axis.

When $x = 0$: $y = \frac{\sqrt{3}}{3} \times 0 + \frac{20\sqrt{3}}{3} = \frac{20\sqrt{3}}{3}$.

So the centre of the circle is $(0, \frac{20\sqrt{3}}{3})$.

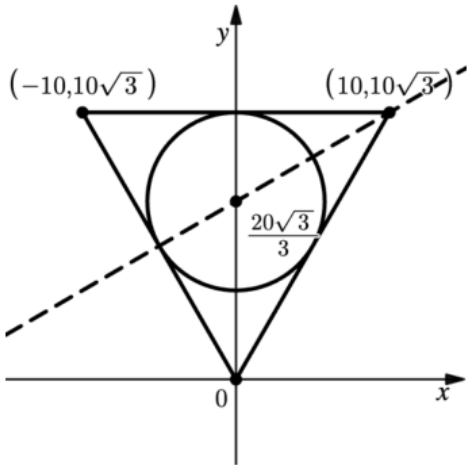
Distance from centre to midpoint of OA:

$r = \sqrt{(-5 - 0)^2 + (5\sqrt{3} - \frac{20\sqrt{3}}{3})^2} = \sqrt{\frac{100}{3}} = \frac{10\sqrt{3}}{3}.$

$\therefore$ The equation of the circle is

$x^2 + (y - \frac{20\sqrt{3}}{3})^2 = \frac{100}{3}.$

The area of the circle is $\pi \times (\sqrt{\frac{100}{3}})^2 = \frac{100\pi}{3}$ units^2.



34)

Centre $= C(3, -5)$.

Gradient of $OC = \frac{-5-0}{3-0} = -\frac{5}{3}$.

Gradient of Tangent $= \frac{3}{5}$.

35)

Centre $= C(-2, 3)$.

Gradient of $AC = \frac{3-2}{-2-1} = \frac{1}{-3} = -\frac{1}{3}$.

Tangent at A: $y - 2 = 3(x - 1) \Rightarrow y = 3x - 1$.

36)

Centre $= C(\frac{1}{2}, -\frac{3}{2})$.

Gradient of $AC = \frac{-\frac{3}{2} + \frac{1}{2}}{\frac{1}{2} - \frac{5}{2}} = \frac{-1}{-2} = \frac{1}{2}$.

Tangent at A: $y + \frac{1}{2} = -2(x - \frac{5}{2}) \Rightarrow y = -2x + \frac{9}{2}$.

37)

Centre $= C(-3, 9)$, Radius $= BC = \sqrt{10}$.

$AC = \sqrt{(1+3)^2 + (1-9)^2} = \sqrt{80}$.

$\angle CBA = 90°$, so AC is the hypotenuse of right-angled triangle ABC.

$\therefore AB^2 = AC^2 - BC^2 = 80 - 10 = 70 \Rightarrow AB = \sqrt{70}$.

38)

Centre $= C(3, 4)$.

When $x = -5$: $(-5-3)^2 + (y-4)^2 = 100$

$\Rightarrow 64 + (y-4)^2 = 100 \Rightarrow (y-4)^2 = 36$

$\Rightarrow y - 4 = \pm 6 \Rightarrow y = \pm 6 + 4 = 10$ or -2.

Let $A(-5, 10)$.

Gradient of $AC = \frac{10-4}{-5-3} = \frac{6}{-8} = -\frac{3}{4}$.

Tangent at A: $y - 10 = \frac{4}{3}(x + 5) \Rightarrow y = \frac{4}{3}x + \frac{50}{3}$.

Let $B(-5, -2)$.

Gradient of $BC = \frac{-2-4}{-5-3} = \frac{-6}{-8} = \frac{3}{4}$.

Tangent at B: $y + 2 = -\frac{4}{3}(x + 5) \Rightarrow y = -\frac{4}{3}x - \frac{26}{3}$.

39)

Centre $= C(-5, 9)$.

When $y = 10$: $(x + 5)^2 + (10 - 9)^2 = 10$

$\Rightarrow (x + 5)^2 + 1 = 10 \Rightarrow (x + 5)^2 = 9$

$\Rightarrow x + 5 = \pm 3 \Rightarrow x = \pm 3 - 5 = -2$ or -8.

Let $A(-2, 10)$.

Gradient of $AC = \frac{10-9}{-2+5} = \frac{1}{3}$.

Tangent at A: $y - 10 = -3(x + 2) \Rightarrow y = -3x + 4$.

Let $B(-8, 10)$.

Gradient of $BC = \frac{10-9}{-8+5} = \frac{1}{-3} = -\frac{1}{3}$.

Tangent at B: $y - 10 = 3(x + 8) \Rightarrow y = 3x + 34$.

40)

$x + 7y = 82 \Rightarrow y = -\frac{1}{7}x + \frac{82}{7}$.

Normal through the centre of the circle:

$y - \frac{17}{2} = 7(x + \frac{5}{2}) \Rightarrow y = 7x + 26$.

The tangent and normal intersect when:

$-\frac{1}{7}x + \frac{82}{7} = 7x + 26 \Rightarrow \frac{50}{7}x = -\frac{100}{7} \Rightarrow x = -2$.

When $x = -2$: $y = 7 \times -2 + 26 = 12$.

So the tangent and normal intersect at $(-2, 12)$.

Radius $= \sqrt{(12 - \frac{17}{2})^2 + (-2 + \frac{5}{2})^2} = \sqrt{\frac{25}{2}}$.

$\therefore (x + \frac{5}{2})^2 + (y - \frac{17}{2})^2 = \frac{25}{2}$.

41 a)

$(x - 11)^2 + (kx + 4)^2 = 16$

$\Rightarrow x^2 - 22x + 121 + k^2x^2 + 8kx + 16 = 16$

$\Rightarrow x^2 + k^2x^2 + 8kx - 22x + 121 = 0$

$\Rightarrow (1 + k^2)x^2 + 2(4k - 11)x + 121 = 0$.

41 b)

If the line is a tangent then $b^2 - 4ac = 0$.

$(2(4k - 11))^2 - 4 \times (1 + k^2) \times 121 = 0$

$\Rightarrow (8k - 22)^2 - 484(1 + k^2) = 0$

$\Rightarrow 64k^2 - 352k + 484 - 484 - 484k^2 = 0$

$\Rightarrow -420k^2 - 352k = 0 \Rightarrow k^2 + \frac{88}{105}k = 0$

$\Rightarrow k(k+\frac{88}{105})=0 \Rightarrow k=0$ or $-\frac{88}{105}$.

$\therefore y=0$ and $y=-\frac{88}{105}x$ are both tangents to the circle.

42 a)

$(x+3)^2+(2x+k-3)^2=80$

$\Rightarrow x^2+6x+9+4x^2+2kx-6x+2kx+k^2-3k-6x-3k+9=80$

$\Rightarrow 5x^2+4kx-6x+k^2-6k-62=0$

$\Rightarrow 5x^2+2(2k-3)x+k^2-6k-62=0$ as required.

42 b)

If the line is a tangent then $b^2-4ac=0$.

$(2(2k-3))^2-4\times 5\times(k^2-6k-62)=0$

$\Rightarrow (4k-6)^2-20(k^2-6k-62)=0$

$\Rightarrow 16k^2-48k+36-20k^2+120k+1240=0$

$\Rightarrow -4k^2+72k+1276=0$

$\Rightarrow k^2-18k-319=0 \Rightarrow (k-29)(k+11)=0$

$\Rightarrow k=29$ or $k=-11$.

$\therefore y=2x+29$ and $y=2x-11$ are both tangents to the circle.

43)

Centre $=C(10,0)$.

When $x=9$: $(9-10)^2+y^2=17$

$\Rightarrow 1+y^2=17 \Rightarrow y^2=16 \Rightarrow y=\pm 4$

$\therefore B(9,4)$.

Gradient of $BC=\frac{4-0}{9-10}=\frac{4}{-1}=-4$.

Tangent at B: $y-4=\frac{1}{4}(x-9) \Rightarrow y=\frac{1}{4}x+\frac{7}{4}$.

When $y=0$: $0=\frac{1}{4}x+\frac{7}{4} \Rightarrow x=-7$

$\therefore A(-7,0)$.

Length of $AB=\sqrt{(9+7)^2+(4-0)^2}=\sqrt{272}$.

Area of Triangle $=\frac{1}{2}\times\sqrt{17}\times\sqrt{272}=34$.

Angle of Sector $=\arccos(\frac{\sqrt{17}}{17})=75.963\ldots^\circ$.

Area of Sector $=\frac{75.963\ldots}{360}\times\pi\times(\sqrt{17})^2=11.269\ldots$

Shaded Area $=34-11.269\ldots=22.7$ units^2.

Alternatively using radians:

Angle of Sector $=\arccos(\frac{\sqrt{17}}{17})=1.325\ldots$

Area of Sector $=\frac{1}{2}\times 1.325\ldots\times(\sqrt{17})^2=11.269\ldots$

Shaded Area $=34-11.269\ldots=22.7$ units^2.

44)

Tangent at A:

$$\begin{aligned}
&(x-25)^2+(-\tfrac{1}{7}x+a-25)^2=1250\\
&\Rightarrow x^2-50x+625+\tfrac{1}{49}x^2-\tfrac{1}{7}ax+\tfrac{25}{7}x-\tfrac{1}{7}ax+a^2-25a+\tfrac{25}{7}x-25a+625=1250\\
&\Rightarrow \tfrac{50}{49}x^2-(\tfrac{300}{7}+\tfrac{2}{7}a)x+a^2-50a=0\\
&\Rightarrow x^2-(42+\tfrac{7}{25}a)x+\tfrac{49}{50}a^2-49a=0.
\end{aligned}$$

Tangent $\Rightarrow b^2-4ac$

$$\begin{aligned}
&(42+\tfrac{7}{25}a)^2-4\times1\times(\tfrac{49}{50}a^2-49a)=0\\
&\Rightarrow 1764+\tfrac{588}{25}a+\tfrac{49}{625}a^2-\tfrac{98}{25}a^2+196a=0\\
&\Rightarrow -\tfrac{2401}{625}a^2+\tfrac{5488}{25}a+1764=0\\
&\Rightarrow a^2-\tfrac{400}{7}a-\tfrac{22500}{49}=0.
\end{aligned}$$

$\Rightarrow a=\frac{450}{7}$, $a\neq-\frac{50}{7}$ as the line crosses through positive y-axis.

$\therefore y=-\frac{1}{7}x+\frac{450}{7}$.

Tangent at B:

$$\begin{aligned}
&(x-25)^2+(-7x+b-25)^2=1250\\
&\Rightarrow x^2-50x+625+49x^2-7bx+175x-7bx+b^2-25b+175x-25b+625=1250\\
&\Rightarrow 50x^2+(300-14b)x+b^2-50b=0\\
&\Rightarrow x^2+(6-\tfrac{7}{25}b)x+\tfrac{1}{50}b^2-b=0.
\end{aligned}$$

Tangent $\Rightarrow b^2-4ac$

$$\begin{aligned}
&(6-\tfrac{7}{25}b)^2-4\times1\times(\tfrac{1}{50}b^2-b)=0\\
&\Rightarrow 36-\tfrac{84}{25}b+\tfrac{49}{625}b^2-\tfrac{2}{25}b^2+4b=0\\
&\Rightarrow -\tfrac{1}{625}b^2+\tfrac{16}{25}b+36=0\\
&\Rightarrow b^2-400b-22500=0.
\end{aligned}$$

$\Rightarrow b=450$, $b\neq-50$ as the line crosses through positive y-axis.

$\therefore y=-7x+450$.

Tangents intersect when $-\frac{1}{7}x+\frac{450}{7}=-7x+450$

$\Rightarrow x-450=49x-3150\Rightarrow 48x=2700\Rightarrow x=\frac{225}{4}$.

$\therefore y=-7\times\frac{225}{4}+450=\frac{225}{4}$. So the tangents intersect at $(\frac{225}{4},\frac{225}{4})$.

45)

Centre $=(0,7)$, Radius $=\sqrt{3}$.

Line passing through $(0,7)$ and perpendicular to $y=2x+1$: $y-7=-\frac{1}{2}(x-0)\Rightarrow y=-\frac{1}{2}x+7$.

$2x+1=-\frac{1}{2}x+7\Rightarrow\frac{5}{2}x=6\Rightarrow x=\frac{12}{5}$.

When $x=\frac{12}{5}$: $y=2\times\frac{12}{5}+1=\frac{29}{5}$.

Distance from $(0,7)$ to $(\frac{12}{5},\frac{29}{5})$:

$=\sqrt{(\frac{12}{5}-0)^2+(\frac{29}{5}-7)^2}=\sqrt{\frac{36}{5}}$.

Distance from Line to Circle is $\sqrt{\frac{36}{5}}-\sqrt{3}=\frac{6\sqrt{5}}{5}-\sqrt{3}$.

46)

$x^2 - 14x + y^2 + 18y - 40 = 0$

$\Rightarrow (x-7)^2 - 49 + (y+9)^2 - 81 - 40 = 0$

$\Rightarrow (x-7)^2 + (y+9)^2 = 170.$

Centre $= C(7, -9)$, Radius $= \sqrt{170}$.

We will call the midpoint of A and B, $M(2, 1)$.

Gradient of $CM = \frac{1+9}{2-7} = \frac{10}{-5} = -2$.

So A and B lie on the line $y - 1 = \frac{1}{2}(x - 2)$

$\Rightarrow y = \frac{1}{2}x.$

$x^2 - 14x + (\frac{1}{2}x)^2 + 18(\frac{1}{2}x) - 40 = 0$

$\Rightarrow x^2 - 14x + \frac{1}{4}x^2 + 9x - 40 = 0$

$\Rightarrow \frac{5}{4}x^2 - 5x - 40 = 0 \Rightarrow x^2 - 4x - 32 = 0$

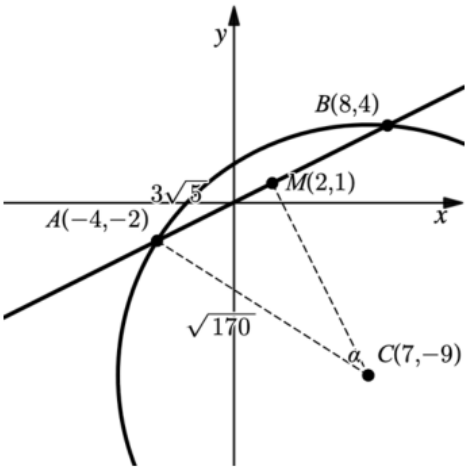
$\Rightarrow (x-8)(x+4) = 0 \Rightarrow x = -4$ or $x = 8$.

When $x = -4$: $y = \frac{1}{2} \times -4 = -2$.

When $x = 8$: $y = \frac{1}{2} \times 8 = 4$.

So $A(-4, -2)$ and $B(8, 4)$.

$AB = \sqrt{(8+4)^2 + (4+2)^2} = \sqrt{180} = 6\sqrt{5}$.



$\alpha = \arcsin(\frac{3\sqrt{5}}{\sqrt{170}}) = 30.963\ldots^\circ$.

Arc Length $= 2 \times \frac{30.963...}{360} \times 2\pi \times \sqrt{170} = 14.1$ units.

47)

Circle P:

$x^2 + y^2 + 6y = 7$ (1)

$\Rightarrow x^2 + (y+3)^2 - 9 = 7$

$\Rightarrow x^2 + (y+3)^2 = 16.$

So circle P has centre $(0, -3)$ and has radius 4.

So we know that circle Q also has radius 4.

Circle Q:

$(x-k)^2+y^2=16$

$\Rightarrow x^2-2kx+k^2+y^2=16$

$\Rightarrow x^2-2kx+y^2=16-k^2$ (2)

(2) − (1): $-2kx-6y=9-k^2$

$\Rightarrow 6y=-2kx-9+k^2$

$\Rightarrow y=-\frac{1}{3}kx-\frac{3}{2}+\frac{1}{6}k^2$.

We will call this line l.

We need l to be a tangent to P, so the line perpendicular to l that passes through the centre of P must also pass through the centre of Q:

$y+3=\frac{3}{k}(x-0)\Rightarrow y=\frac{3}{k}x-3$.

We will call this line m.

Find where l and m intersect:

$-\frac{1}{3}kx-\frac{3}{2}+\frac{1}{6}k^2=\frac{3}{k}x-3\Rightarrow\frac{3}{2}+\frac{1}{6}k^2=\frac{3}{k}x+\frac{1}{3}kx$

$\Rightarrow(\frac{3}{k}+\frac{1}{3}k)x=\frac{3}{2}+\frac{1}{6}k^2$

$$\Rightarrow x=\frac{\frac{3}{2}+\frac{1}{6}k^2}{\frac{3}{k}+\frac{1}{3}k}\equiv\frac{\frac{3}{2}k+\frac{1}{6}k^3}{3+\frac{1}{3}k^2}\equiv\frac{9k+k^3}{18+2k^2}\equiv\frac{k}{2}\times\frac{9+k^2}{9+k^2}\equiv\frac{k}{2}.$$

When $x=\frac{k}{2}$: $y=\frac{3}{k}\times\frac{k}{2}-3=-\frac{3}{2}$.

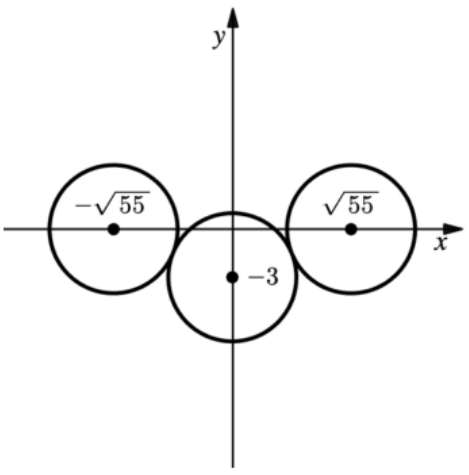
So the circles intersect at $(\frac{k}{2},-\frac{3}{2})$.

The distance from $(k,0)$ to $(\frac{k}{2},-\frac{3}{2})$ must be 4:

$\sqrt{(\frac{k}{2}-k)^2+(-\frac{3}{2}-0)^2}=4\Rightarrow\sqrt{\frac{k^2}{4}+\frac{9}{4}}=4$

$\Rightarrow\frac{k^2}{4}+\frac{9}{4}=16\Rightarrow k^2=55\Rightarrow k=\pm\sqrt{55}$.

So Q has equation $(x-\sqrt{55})^2+y^2=16$ or $(x+\sqrt{55})^2+y^2=16$.



48)

$3x-2y=12\Rightarrow y=\frac{3}{2}x-6$.

We can use $y=-\frac{2}{3}x$ to calculate the diameter of the circle:

$3x-2(-\frac{2}{3}x)=12\Rightarrow\frac{13}{3}x=12\Rightarrow x=\frac{36}{13}$.

When $x = \frac{36}{13}$: $y = -\frac{2}{3} \times \frac{36}{13} = -\frac{24}{13}$.

$3x - 2(-\frac{2}{3}x) = 24 \Rightarrow \frac{13}{3}x = 24 \Rightarrow x = \frac{72}{13}c$.

When $x = \frac{72}{13}$: $y = -\frac{2}{3} \times \frac{72}{13} = -\frac{48}{13}$.

Distance between $(\frac{36}{13}, -\frac{24}{13})$ and $(\frac{72}{13}, -\frac{48}{13})$ is

$= \sqrt{(\frac{72}{13} - \frac{36}{13})^2 + (-\frac{48}{13} + \frac{24}{13})^2} = \sqrt{\frac{144}{13}}$.

So Radius $= \frac{1}{2}\sqrt{\frac{144}{13}} = \frac{6\sqrt{13}}{13}$.

$\therefore (x-8)^2 + (y-k)^2 = \frac{36}{13}$.

$$
\begin{aligned}
&(x-8)^2 + (\tfrac{3}{2}x - 6 - k)^2 = \tfrac{36}{13} \\
&\Rightarrow x^2 - 16x + 64 + \tfrac{9}{4}x^2 - 9x - \tfrac{3}{2}kx - 9x + 36 + 6k - \tfrac{3}{2}kx + 6k + k^2 = \tfrac{36}{13} \\
&\Rightarrow \tfrac{13}{4}x^2 - 3kx - 34x + k^2 + 12k + \tfrac{1264}{13} = 0 \\
&\Rightarrow \tfrac{13}{4}x^2 - (3k + 34)x + k^2 + 12k + \tfrac{1264}{13} = 0.
\end{aligned}
$$

Tangent $\Rightarrow b^2 - 4ac = 0$

$(3k+34)^2 - 4 \times \frac{13}{4} \times (k^2 + 12k + \frac{1264}{13}) = 0$

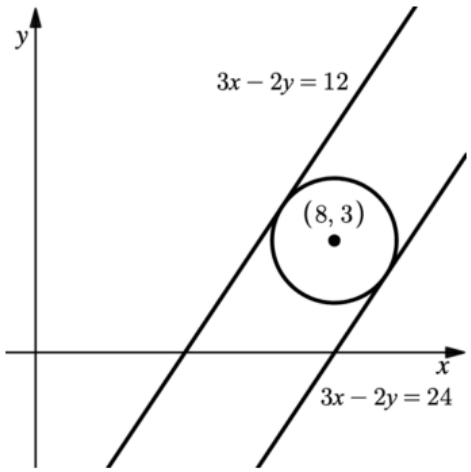
$\Rightarrow 9k^2 + 204k + 1156 - 13k^2 - 156k - 1264 = 0$

$\Rightarrow -4k^2 + 48k - 108 = 0 \Rightarrow k^2 - 12k + 27 = 0$

$\Rightarrow (k-3)(k-9) = 0 \Rightarrow k = 3$ or $k = 9$.

$k = 3$ as we want the circle to be underneath $3x - 2y = 12$.

$\therefore (x-8)^2 + (y-3)^2 = \frac{36}{13}$.



49)

The circle has equation $(x-k)^2+(y-k)^2=k^2$.

$2x+y=50 \Rightarrow y=50-2x$.

$$\begin{aligned}
&(x-k)^2+(50-2x-k)^2=k^2\\
&\Rightarrow x^2-2kx+k^2+2500-100x-50k-100x+4x^2+2kx-50k+2kx+k^2=k^2\\
&\Rightarrow 5x^2+2kx-200x+k^2-100k+2500=0\\
&\Rightarrow 5x^2+(2k-200)x+k^2-100k+2500=0.
\end{aligned}$$

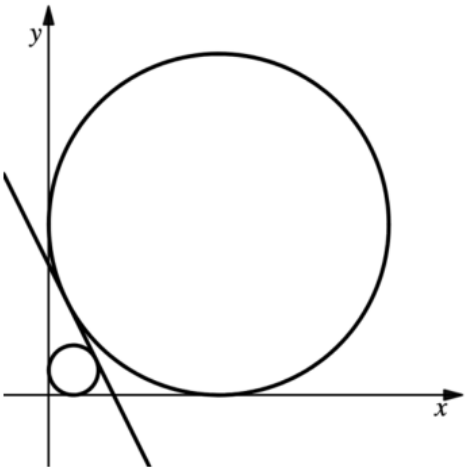
Tangent $\Rightarrow b^2-4ac=0$

$(2k-200)^2-4\times 5\times(k^2-100k+2500)=0$

$\Rightarrow 4k^2-800k+40000-20k^2+2000k-50000=0$

$\Rightarrow -16k^2+1200k-10000=0$

$\Rightarrow k^2-75k+625=0 \Rightarrow k=\frac{75\pm 25\sqrt{5}}{2}$.



50)

Area $= 25\pi = \pi r^2 \Rightarrow r = 5$.

So the centre has coordinates $(k, 5)$ and the equation of the circle is $(x-k)^2 + (y-5)^2 = 25$.

Sub in $y = \frac{1}{2}x$: $(x-k)^2 + (\frac{1}{2}x - 5)^2 = 25$

$\Rightarrow x^2 - 2kx + k^2 + \frac{1}{4}x^2 - 5x + 25 = 25$

$\Rightarrow \frac{5}{4}x^2 - 2kx - 5x + k^2 = 0$

$\Rightarrow \frac{5}{4}x^2 - (2k+5)x + k^2 = 0.$

Tangent $\Rightarrow b^2 - 4ac$

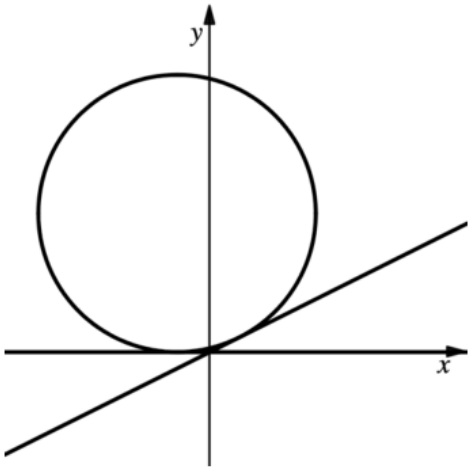
$(2k+5)^2 - 4 \times \frac{5}{4} \times k^2 = 0$

$\Rightarrow 4k^2 + 20k + 25 - 5k^2 = 0$

$\Rightarrow -k^2 + 20k + 25 = 0$

$\Rightarrow k^2 - 20k - 25 = 0 \Rightarrow k = 10 \pm 5\sqrt{5}.$

$k = 10 + 5\sqrt{5}$ as $k = 10 - 5\sqrt{5}$ gives:



$\therefore (x - 10 - 5\sqrt{5})^2 + (y-5)^2 = 25.$

51)

Radius $= \sqrt{(2-1)^2 + (2+\sqrt{7}-2)^2} = \sqrt{8}$.

So the equation of the circle is

$(x-1)^2 + (y-2)^2 = 8$.

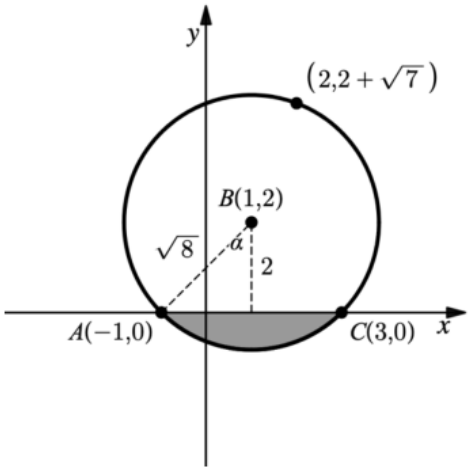
$y = 0$: $(x-1)^2 + (-2)^2 = 8 \Rightarrow (x-1)^2 + 4 = 8$

$\Rightarrow (x-1)^2 = 4 \Rightarrow x - 1 = \pm 2$

$\Rightarrow x = \pm 2 + 1 = 3$ or -1.

So the circle intersects the x-axis at $(-1, 0)$ and $(3, 0)$.

We will label $A(-1, 0)$, $B(1, 2)$ and $C(3, 0)$.



We now need to find the $\angle ABC$.

$\alpha = \arccos(\frac{2}{\sqrt{8}}) = 45°$ so $\angle ABC = 90°$.

Area $= \frac{1}{4} \times \pi \times (\sqrt{8})^2 - \frac{1}{2} \times \sqrt{8} \times \sqrt{8} = 2\pi - 4$.

Percentage shaded $= \dfrac{2\pi - 4}{\pi \times (\sqrt{8})^2} \times 100 = 9.08\%$ to 3sf.